

MATTHEW A. COTAJ

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EDUCATION

Georgetown University

Washington, DC · May 2026

Bachelor of Science in Chemistry (Honors), Minor in Biology

GPA: 3.5 / 4.0

University of New South Wales (UNSW)

Sydney, Australia · Jan – May 2025

Study Abroad · Coursework: Materials Chemistry, Physics

EXPERIENCE

Pfizer

Groton, CT · June 2026 – Present

Biopharmaceutics Intern, Drug Product Design (Pfizer Futures Program)

- Characterized V1/V2 biorelevant media across 20+ conditions via DLS and tensiometry, standardizing prep across 4 functions
- Correlated colloidal media properties to solubility and dissolution of 5 model compounds, improving in-vitro to in-vivo translation ~30%
- Identified surfactant-bile micelle interactions forming solubility-limiting complexes, flagging excipient risks in early-phase formulations

Stoll Research Group, Georgetown University

Washington, DC · Feb 2023 – May 2026

Honors Thesis Candidate & Materials Science Undergraduate Research Fellow

- Achieved the first colloidal synthesis and magnetic characterization of phase-pure EuGdCuS₃ nanorods using single-source dithiocarbamate precursors under air-free conditions (Schlenk line, glovebox)
- Demonstrated compositional control of magnetism by substituting Gd³⁺ for Eu²⁺, raising saturation magnetization from 47.8 to ~78 emu/g
- Screened lanthanide halide and sulfur precursors to phase-control binary (Nd₃S₄, Gd₃S₄) and ternary nanomaterials, verifying composition and morphology by PXRD, ICP-MS, EDS, and HRTEM

Georgetown University, Department of Chemistry

Washington, DC · Aug 2025 – May 2026

General Chemistry I & II Lab Teaching Assistant

- Guided students through general chemistry experiments, reinforcing safe and accurate lab technique
- Set up and maintained lab equipment and reagents across weekly lab sections
- Supported instructors in running sessions and resolving student questions

Henkel

Stamford, CT · May – Aug 2025

Product Development Intern

- Developed and validated a method to differentiate liquid soap formulations for a \$440M product line
- Optimized formulation variables, improving foam performance 23% across all Dial products
- Supported antibacterial and non-antibacterial soap development under FDA and OTC compliance

CG Pharmaceuticals

Washington, DC · Mar – July 2024

Rare Disease Therapeutics Research Intern

- Researched alternative treatments for neurodegenerative disorders (Friedreich's Ataxia and Fragile X Syndrome)
- Conducted scientific and IP research to support seed-stage biotech strategy
- Assessed drug targets and biological pathways, evaluating feasibility on IP ownership and clinical relevance

ANAMA Package & Container Testing Services

Stamford, CT · Summers 2022–2024

Project Manager Intern | Package Testing Technician

- Developed and executed testing protocols ensuring industry-standard packaging compliance
- Performed UN/USDOT/ISTA testing for clients including Bloomberg, Duracell, and the U.S. Army
- Managed resources and budgets to double per-diem project completion

PUBLICATIONS & PRESENTATIONS

Hartman, T.; **Cotaj, M.**; El Jamal, E.S.; Stoll, S.L. "Magnetic and magneto-optical properties of mixed cation and mixed valent EuLnCuQ₃ (Q = S, Se) nanorods." *Manuscript in preparation*.

Henkel Executive R&D Board Meeting, Trumbull, CT (Aug 2025) — presented next-generation foaming chassis strategy for Dial and Henkel systems.

TECHNICAL SKILLS

Materials & Analytical: PXRD (Ultima IV), SQUID, SEM, TEM, XPS, EDX, AFM, ICP-OES, UPLC, HPLC, NMR, IR/FTIR, UV-Vis, Raman, DLS, LC-MS, GC-MS, cyclic voltammetry, surface plasmon resonance, CVD, sputtering, thin-film deposition, etching, photolithography

Computational: Python, R, OriginLab, JMP, ChemDraw, Gaussian, Spartan, CSD, MDI Jade, Snowflake, SpinWorks

Trainings: GxPs, HAZMAT, Highly Potent Compounds (HPCs), Laser, Radioactive Materials (RAM)